

VERTEX SYSTEMS INC.

Fictional Company · Sample Job Description

Backend Software Engineer (Python / Go)

Location: Ahmedabad, India (Hybrid) | Department: Core Engineering | Type: Full-time | Experience: 0-2 years

ABOUT THE ROLE

Vertex Systems Inc. builds the transaction processing backbone used by fintech and logistics clients to handle millions of daily events with sub-second latency guarantees. We're hiring a Backend Software Engineer to join our Core Engineering team, working on high-throughput APIs, asynchronous job processing, and the data layer that powers our platform. This role is ideal for an engineer who enjoys reasoning about performance, correctness, and clean system design.

KEY RESPONSIBILITIES

- Design, build, and maintain RESTful and gRPC APIs serving both internal services and external partners.
- Write efficient, well-tested backend services in Python (FastAPI) and Go for latency-sensitive workloads.
- Model and optimize relational schemas in PostgreSQL, including indexing strategy and query performance tuning.
- Build asynchronous processing pipelines using message queues (Redis Streams / Kafka) for background workloads.
- Implement caching layers and rate-limiting strategies to protect downstream systems under load.
- Write unit, integration, and load tests; participate actively in code review to uphold engineering standards.
- Collaborate with the platform team to containerize services and integrate with CI/CD pipelines.
- Debug production incidents using logs, metrics, and distributed tracing tools.

MUST-HAVE QUALIFICATIONS

- 0-2 years of professional or internship experience in backend software development.
- Strong fundamentals in data structures, algorithms, and object-oriented design.
- Working proficiency in Python and at least familiarity with a statically typed language (Go, Java, or similar).
- Understanding of relational databases (PostgreSQL/MySQL) including schema design and SQL query writing.
- Basic understanding of RESTful API design principles and HTTP semantics.
- Familiarity with Git-based workflows and collaborative code review practices.
- Ability to write clear, maintainable code with adequate test coverage.

NICE TO HAVE

- Exposure to Go for building backend services or CLI tooling.
- Experience with message brokers such as Kafka, RabbitMQ, or Redis Streams.
- Understanding of containerization (Docker) and basic Kubernetes concepts.
- Contributions to open-source projects or personal projects published on GitHub/PyPI.
- Exposure to system design concepts: caching, sharding, load balancing.

BENEFITS & PERKS

- Structured mentorship program pairing new hires with senior engineers.
- Hybrid work model with 2 in-office collaboration days per week.
- Annual learning budget for courses, books, and technical certifications.
- Comprehensive health and accident insurance.
- Performance-linked bonus reviewed twice a year.

This is a fictional job posting created for practice/demonstration purposes only. Any resemblance to real companies or postings is coincidental.